AITS
ALL INDIA TEST SERIES

11th JEE Main

AJAM7/06 **Test- 06**

DURATION : 180 Minutes **DATE : 23/02/2025** **M.MARKS : 300**

ANSWER KEY

PHYSICS

1. (2)
2. (1)
3. (1)
4. (3)
5. (2)
6. (1)
7. (1)
8. (3)
9. (4)
10. (1)
11. (2)
12. (4)
13. (1)
14. (1)
15. (3)
16. (4)
17. (2)
18. (3)
19. (1)
20. (2)
21. (7)
22. (25)
23. (10)
24. (10)
25. (360)

CHEMISTRY

26. (2)
27. (3)
28. (1)
29. (1)
30. (2)
31. (3)
32. (3)
33. (4)
34. (3)
35. (3)
36. (2)
37. (3)
38. (2)
39. (2)
40. (2)
41. (3)
42. (4)
43. (2)
44. (1)
45. (2)
46. (6)
47. (11)
48. (7)
49. (29)
50. (12)

MATHEMATICS

51. (1)
52. (3)
53. (3)
54. (3)
55. (2)
56. (3)
57. (3)
58. (2)
59. (1)
60. (4)
61. (3)
62. (4)
63. (3)
64. (1)
65. (2)
66. (4)
67. (3)
68. (1)
69. (1)
70. (3)
71. (14)
72. (10)
73. (3)
74. (25)
75. (135)

SECTION-I (PHYSICS)

1. **(2)**

Initially, $F = \frac{Gm^2}{r^2}$ …(i)

When Δm mass of the object is transferred to the other, then now force

$$F' = \frac{G \times (m - \Delta m)(m + \Delta m)}{\frac{4r^2}{9}} = \frac{8F}{9} \quad \text{…(ii)}$$

$$\frac{9}{4}\frac{G\left(m^2 - \Delta m^2\right)}{r^2} = \frac{Gm^2}{r^2} \times \frac{8}{9}$$

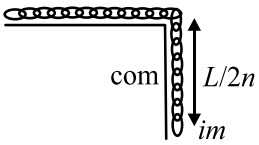
$$\Rightarrow \quad 81\left(m^2 - \Delta m^2\right) = 32m^2$$

$$\Rightarrow \quad \frac{\Delta m}{m} = \frac{7}{9}$$

[28 June, 2022 (Shift-II)]

2. **(1)**

Let the length of portion of rope hanging is $\frac{L}{n}$



$$mg\frac{L}{n} + \frac{M}{n}g\frac{L}{2n}$$

Work done, $W = mgh_{COM}$

$$\Rightarrow \quad \frac{MgL}{g} = \frac{MgL}{6n} + \frac{MgL}{2n^2}$$

$$\Rightarrow \quad 2n^2 - 3n - 9 = 0$$

$$\Rightarrow \quad n = 3$$

Length of having portion of rope is

[9 April, 2019 (Shift-I)]

3. **(1)**

For this process, Q = 0, W = 0, ΔU = 0, and therefore (since it is an ideal gas) ΔT = 0. One might think that the entropy change is zero because there is not heat exchange. This free expansion is not reversible, and there is an entropy change.

4. **(3)**

For 16 kg block

$16g - T = 16a$ …(i)

For 24 kg block

$T - 0.5 \times 24\, g = 24\, a$ …(ii)

Adding (i) and (ii), we get

$a = 1 \text{m/s}^2$

$\Rightarrow \quad T = 16(10 - 1) = 144$ N

Clamp force $= T\sqrt{2} = 144\sqrt{2} \approx 204$ N

[29 June, 2022 (Shift-II)]

5. **(2)**

$$\alpha\beta = \left[\frac{MLT^{-2}}{L^2}\right]\left[\frac{MLT^{-2}}{L}\right]$$

$$= \left[M^2L^{-1}T^{-4}\right]$$

[27 July, 2022 (Shift-II)]

6. **(1)**

By using Hooke's Law

$F \propto \Delta L$

$$\frac{F_1}{F_2} = \frac{\Delta L_1}{\Delta L_2}$$

$$\Rightarrow \frac{10}{20} = \frac{x_1 - x}{x_2 - x}$$

$\Rightarrow x = 2x_1 - x_2,$

$$\text{Stress } 10kg = \left(\frac{x_1 - x}{x}\right)Y$$

$$\text{Stress } 10kg = \left(\frac{x_2 - x_1}{2x_1 - x_2}\right)Y$$

[29 June, 2022 (Shift-I)]

7. **(1)**

$$v^2 = u^2 + 2gh$$

$$\Rightarrow \quad u^2 = v^2 - 2gh$$

$$\Rightarrow \quad u^2 = 25^2 - 2 \times 10 \times 11.25 = 400$$

$$\Rightarrow \quad u = 20 \text{ m/s}$$

[1 Feb, 2023 (Shift-I)]

8. **(3)**

$$v_{\max} = \sqrt{\left(\frac{\tan\theta + \mu}{1 - \mu + \tan\theta}\right)rg}$$

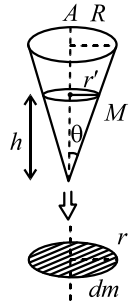
On solving we get $\mu = \frac{3}{5} = 0.6$

[29 Jan, 2023 (Shift-I)]

9. **(4)**

The moment of inertia of a solid cone mass *M*, radius *R* is

$$= \frac{3MR^2}{10}$$



When cut and mid-point

$r' = \left(\frac{R}{2}\right)$

$h' = \left(\frac{R}{2}\right)$

Mass of lower cone = $\frac{M}{8}$

$M.I. = \left(\frac{3}{320}\right)MR^2$

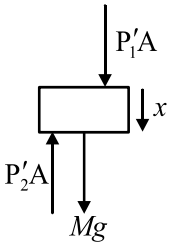
$M.I.$ of frustum = $\frac{3MR^2}{10} - \frac{3MR^2}{320} = \frac{93}{320}MR^2$

[6 Sep, 2020 (Shift-I)]

10. (1)

As piston is stationary. So using *FBD* of the piston,

$P_2A - P_1A = mg$



$\left(P_1' - P_2'\right)A - mg = ma$ ….(i)

Using PV^γ = constant

$P_1\left(Al_1\right)^\gamma = P_1'\left(A\left(l_1 + x\right)\right)^\gamma$ …(ii)

$P_2\left(Al_2\right)^\gamma = P_2'\left(A\left(l_2 - x\right)\right)^\gamma$ ….(iii)

Using equation (i), (ii) and (iii)

We have

$a = -\left[\frac{\gamma\left(l_1^2 + l_2^2\right)nRT}{l_1^2 l_2^2 m}\right]x$

$T = 2\pi l_1 l_2\sqrt{\frac{m}{\gamma\left(l_1^2 + l_2^2\right)nRT}}$

[12 Jan, 2019 (Shift-II)]

11. (2)

$a = \frac{5g}{10} = \frac{g}{2}$

Tension in string joining 6th and 7th balls is

$T = 4g = 40$ N

[26 June, 2022 (Shift-II)]

12. (4)

Given that, $\vec{A}.\vec{B} = \left|\vec{A} \times \vec{B}\right| \Rightarrow AB\cos\theta = AB\sin\theta$

$\Rightarrow \theta = 45°$

$A^2 - B^2 = 0$

$A = B$

$\left|\vec{A} + \vec{B}\right| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$= \sqrt{A^2 + B^2 + 2AB \times \frac{1}{\sqrt{2}}} = \sqrt{A^2 + B^2 + \sqrt{2}AB}$

$\left|\vec{A} + \vec{B}\right| = \sqrt{A^2 + A^2 + 2AA\cos\theta}$

$= A\sqrt{2 + \sqrt{2}}$

[20 July, 2021 (Shift-I)]

13. (1)

$f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\ell}\sqrt{\frac{\sigma}{\rho}} = \frac{1}{2\ell}\sqrt{\frac{Y\varepsilon}{\rho}}$

14. (1)

$\because$ Zero Error = 8 × Least Count = 0.4 mm

$\therefore$ Reading = $MSR + VSR \times LC$ – Zero Error

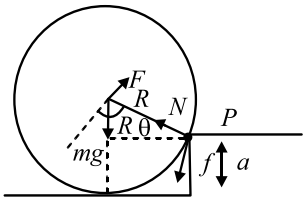
= [42 mm + (0.05) 6mm] – 0.4 mm

= 41.9 mm = 41.9×10^{-1} cm = 4.19 cm

= 4.19 cm

[10 Apr, 2023 (Shift-II)]

15. (3)



$FR > mg\cos\theta\, R$

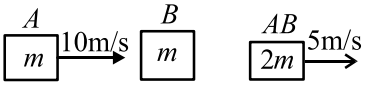
$F > mg\cos\theta$

$F > mg\sqrt{\frac{R^2 - (R-a)^2}{R^2}} \Rightarrow Mg\sqrt{1 - \frac{(R-a)^2}{R^2}}$

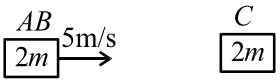
$F = 50\sqrt{1 - \left(\frac{1}{2}\right)^2} = 25\sqrt{3}$

[2 Sep, 2020 (Shift-I)]

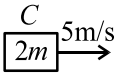
16. (4)



After collision of *A* and *B*



After collision of *AB* and *C*, *AB* comes to rest



[27 July, 2021 (Shift-I)]

17. (2)

$v = \left(\sqrt{t}\hat{i} + t^3\hat{j}\right)$

$a = \frac{dv}{dt} = \frac{1}{2\sqrt{t}}\hat{i} + 3t^2\hat{j}$

When $t = 1\text{sec} \Rightarrow a = \frac{1}{2}\hat{i} + 3\hat{j}$

$\therefore\ F = ma = 2\left(\frac{1}{2}\hat{i} + 3\hat{j}\right)$

$F = \sqrt{37}$ N

18. (3)

$$\sqrt{\frac{2l}{a_2}} - \sqrt{\frac{2l}{a_1}} = t \Rightarrow \frac{\sqrt{2l}}{t} - \frac{\sqrt{a_1 a_2}}{\sqrt{a_1} - \sqrt{a_2}}$$

$$\sqrt{2a_1 l} - \sqrt{2a_2 l} = v \Rightarrow \frac{\sqrt{2l}}{v} = \frac{1}{\sqrt{a_1} - \sqrt{a_2}}$$

$$\Rightarrow \frac{v}{t} = \sqrt{a_1 a_2} \Rightarrow v = \left(\sqrt{a_1 a_2}\right)t$$

t = 10s, a_1 = 4 m/s^2, a_2 = 2 m/s^2

$v = 20\sqrt{2}$ m/s^2

[9 Jan, 2019 (Shift-II)]

19. (1)

$$\rho(r) = \left(\frac{K}{r}\right)$$

$$m = \int_0^r \left(\frac{K}{r}\right) \times 4\pi r^2 dr$$

$$E_g = \frac{Gm}{r^2}$$

$$E_g = 2\pi\ KG$$

$$m' E_g = \frac{m' V^2}{r}$$

$$v = \sqrt{2\pi KGr}$$

$$T = \frac{2\pi r}{v}$$

$$T = \sqrt{\frac{2\pi r}{KG}}$$

$$\frac{T}{\sqrt{r}} = \text{constant}$$

[9 April, 2019 (Shift-II)]

20. (2)

For combined motion

$F - 20 = 5a$ ……(i)

a_{max} for combined motion = 0.6 g = 6 m/s^2

F_{max} = 20 + 30 = 50 N

[10 April, 2019 (Shift-II)]

21. (7)

$$K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}} = \frac{6+3}{\frac{6}{K} + \frac{3}{3K}}$$

$$K_{eq} = \frac{9K}{7} = \left(1 + \frac{2}{7}\right)K$$

$\alpha = 7$

[29 June, 2022 (Shift-I)]

22. (25)

$W = nR\Delta T$

$\Delta Q = \Delta U + W$

$= \frac{f}{2} nR\Delta T + nR\Delta T$

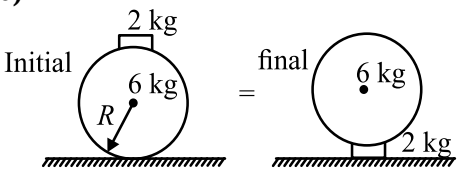
100 = (3 + 1) $nR\Delta T$

$\Rightarrow \quad nR\Delta T$ = 25

$W = nR\Delta T$ = 25 J

[28 July, 2022 (Shift-II)]

23. (10)



From conservation of energy.

Gain in K.E = Loss in P.E

$$mg(2R) = \frac{1}{2} I_d \omega^2 + \frac{1}{2} I_p \omega^2$$

or $\quad 2 \times 2 \times 10 \times 0.1 = = \frac{1}{2} \times (6+2) v^2$

$\omega^2 R^2 = 1$

ω^2 = 100 rad^2 s^{-2}

ω = 10 rad s^{-1}

[26 July, 2022 (Shift-I)]

24. (10)

Displacement, $x = A \sin(\omega t + \phi)$

Velocity, $= A\omega \cos(\omega t + \phi)$

At $t = 0$

$\Rightarrow 3 = A \sin\phi$ …(i)

$\omega = A\omega \cos\phi$ …(ii)

Solving (i) and (ii) we get

$A^2 = 10$

$A = \sqrt{10}$ cm

[27 July, 2021 (Shift-II)]

25. (360)

$X_1 = 10 - 3t^2$

$\vec{v}_1 = -6t\hat{i}$

$Y_2 = 5 - 6t + 8t^3$

$\vec{v}_2 = \left(-6 + 24t^2\right)\hat{j}$

at t = 1sec

$= \left|\vec{v}_2 - \vec{v}_1\right| = \left|-18\hat{j} - 6\hat{i}\right|$

$v = \sqrt{18^2 + 6^2}$

$v = \sqrt{360}$ m/s

[8 Jan, 2020 (Shift-I)]

SECTION-II (CHEMISTRY)

26. (2)

Elements in the same period have different chemical properties because they have different numbers of valence electrons. Chemical properties are more similar within a group, not within a period.

[22 July, 2021 (Shift-II)]

27. (3)

$K_w = \left[H^+\right]\left[OH^-\right]$

With increase in temperature, K_w increases.

For pure water $\left[H^+\right] = \left[OH^-\right]$

$K_w = \left[H^+\right]^2 \Rightarrow \left[H^+\right] = \sqrt{K_w}$

On increasing temperature, $K_w \uparrow \left[H^+\right] \uparrow$ pH $\downarrow$

Dissociation of H_2O into H^+ and OH^- is endothermic (ΔH is +ve)

[8 Jan, 2020 (Shift-II)]

28. (1)

$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$ is a disproportionation reaction. oxidation state of Mn is changing from +6 to +7 and +4.

[12 April, 2019 (Shift-I)]

29. (1)

The given molecules can be categorised as:

BF_3 – electron deficient molecule

NO_2 – odd electron

[29 July, 2022 (Shift-I)]

30. (2)

Moles of $= \frac{15}{75} = 0.2$ mol

$X \rightleftharpoons Y(g) + Z(g)$

0.2	_	_	Initially
$0.2(1-\alpha)$	0.2α	0.2α	At eq.

Given α 25% = 0.25

Therefore number of moles equilibrium due

$0.2 \times 0.25 + 0.2 \times 0.25$

$0.05 + 0.05 = 0.1$

As we know PV = nRT

At equlibrium,

$P_{total} \times 5 = 0.1 \times .0821 \times 573$

$P_{total} = 0.940866$

At equilibrium,

$P_y = P_z = \frac{P_{total}}{2} = \frac{0.940866}{2} = 0.4704$

$K_p = P_y \times P_z$

$K_p = 0.2213 atm^2$

[10 Jan, 2019 (Shift-II)]

31. (3)

$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O$

15 $\quad 15\left(x + \frac{y}{4}\right) \quad 15x$

By given data, $15\left(x + \frac{y}{4}\right) = 75$ … (i)

$15x = 60$ … (ii)

$\therefore \quad x = 4, y = 4$

[5 Sept, 2020 (Shift-I)]

32. (3)

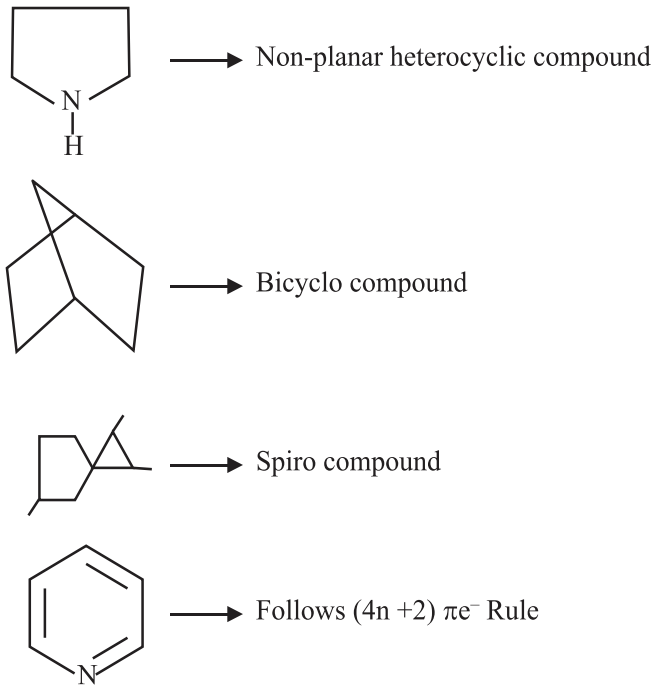
According to the electromagnetic spectrum, Paschen series appears in the infrared region**.**

33. (4)

The outer most electronic configuration of Tl is $6s^2 6p^1$. 6s electrons are strongly attracted towards the nucleus due to its more penetrating power and deshielding of d and f-electrons, Hence, 6s-electrons do not participate in bonding

[9 Jan, 2019 (Shift-I)]

34. (3)

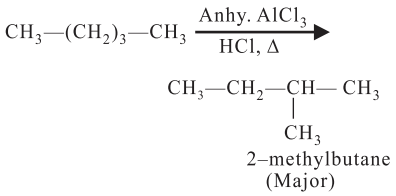


[28 July, 2022 (Shift-I)]

35. (3)

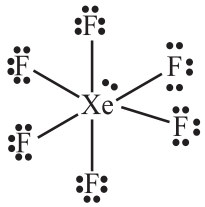
When heated in the presence of hydrogen chloride gas and anhydrous $AlCl_3$, n-alkanes are isomerized to branched chain alkanes. The primary product has one side chain of methyl.

The reaction takes place as:



[13 April, 2023 (Shift-I)]

36. (2)



37. (3)

(Bond enthalpy order C – C > Si – Si > Ge – Ge > Sn – Sn) As we move down the group, bond length increases and bond enthalphy decreases.

Bond enthalpy order S – S > O – O is due to interelectronic repulsion which is absent in 14 group elements

38. (2)

Conceptual

[3 Sept, 2020 (Shift-I)]

39. (2)

The reaction for a diprotic acid (H_2X) with NaOH is:

$H_2X + NaOH \rightarrow NaHX + H_2O$

$NaHX + NaOH \rightarrow Na_2X + H_2O$

Therefore, 10 moles of H_2X are required.

[25 July, 2022 (Shift-I)]

40. (2)

α-particles passing through with minimal deflection suggests that the positive charge is distributed evenly and does not create a significant localized charge to repel the α-particles strongly.

[27 July, 2021 (Shift-II)]

41. (3)

The correct match for given column I and column II is:

(A) - (Q), (B) - (R), (C) - (S), (D) - (P).

[27 July, 2022 (Shift-II)]

42. (4)

- For an ideal gas, the internal energy U depends solely on the temperature.
- For an ideal gas, C_V is a function of the number of degrees of freedom and is independent of volume.
- For a monatomic ideal gas, γ (the ratio of specific heats) is 5/3, not 1.
- For an ideal gas, $C_P > C_V$ because $C_P = C_V + nR$, where R is the gas constant and n is the number of moles. This statement is correct.

[12 Jan, 2019 (Shift-I)]

43. (2)

$Ca_3(PO_4)_2 \rightarrow 3Ca^{2+} + 2PO_4^{3-}$

1 mole of $Ca_3(PO_4)_2$ dissociates to produce 3 moles of Ca^{2+} and 2 moles of PO_4^{3-}

Let the solubility of $Ca_3(PO_4)_2$ S mol L^{-1}

Given S = 1.0×10^{-7} S mol L^{-1}

The concentration of Ca^{2+} ions will be 3S

The concentration of PO_4^{3-} ions will be 2S.

$\left[PO_4^{3-}\right] = 2.0 \times 10^{-7}\ \text{mol}L^{-1}$

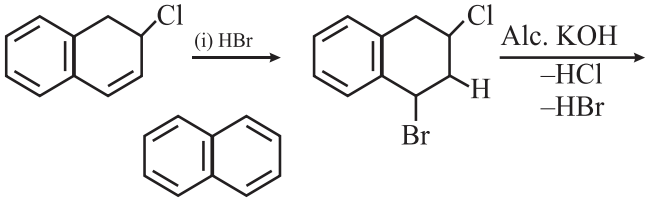
[25 June, 2022 (Shift-I)]

44. (1)

As in HNO_3, nitrogen is present in it's maximum oxidation state hence it cannot act as reducing agent.

[9 Jan, 2020 (Shift-I)

45. (2)



[9 Jan, 2019 (Shift-I)]

46. (6)

Density: Intensive

Heat capacity: Extensive

Molarity: Intensive

Standard cell potential ($E°_{cell}$): Intensive

Entropy: Extensive

Specific volume: Intensive

Molar mass: Intensive

Boiling point: Intensive

Refractive index: Intensive

[11 April, 2023 (Shift-II)]

47. (11)

un stands for 1, quad for 4

So, atomic number = 114

x = 114

$\frac{x}{10} = 11.4$

[3 Sept, 2020 Shift-I)]

48. (7)

$A(g) \rightleftharpoons 2B(g) + C(g)$

At t = 0 500

At time, t 500 – x 2x x

Now, $P_T = P_A + P_B + P_C$ (Given: P_T = 800 mm Hg)

800 = 500 – x + 2x + x

2x = 300

x = 150

Hence, the fraction of A decompose

$\frac{150}{500} = 0.3$

The fraction of A remaining = 1 – 0.3 = 0.7

a = 0.7

10a = 7

[10 April, 2023 (Shift-II)]

49. (29)

: PV = nRT

$$n = \frac{(1.2 \text{ atm}) \times (0.320 \text{ L})}{(0.0821 \text{ L atm mol}^{-1}\text{K}^{-1}) \times (303.15 \text{ K})}$$

$n \approx 0.0154$ mol

Mass of N_2 = n × molar mass Mass of N_2 = 0.0154 mol × 28 g/mol

Mass of $N_2 \approx 0.432$g

$$\text{Percentage of N} = \left(\frac{\text{Mass of } N_2}{\text{Mass of sample}}\right) \times 100$$

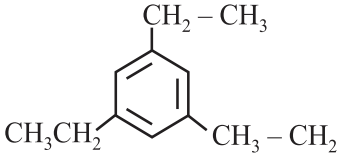
Percentage of N = [0.432 g/1.50 g] × 100%

Percentage of N ≈ 28.8%

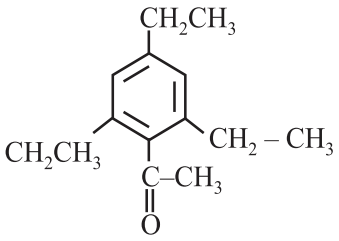
[27 Aug, 2021 (Shift-I)]

50. (12)

A:



B:



[8 Jan, 2020 (Shift-II)]

SECTION-III (MATHEMATICS)

51. (1)

1.	**2.**	**3.**	**4.**	**5.**
6.	**7.**	**8.**	**9.**	**10.**
11.	**12.**	**13.**	**14.**	**15.**
16.	**17.**	**18.**	**19.**	**20.**
21.	**22.**	**23.**	**24.**	**25.**

So, required numbers = $5 \times {}^4C_3 \times 3! = 120$

[12 April, 2023 (Shift-I)]

52. (3)

$A = \{x \in R : [x + 3] + [x + 4] \le 5\}$

$[x] + 3 + [x] + 4 \le 5$

$2[x] \le -2$

$[x] \le -1 \Rightarrow x, < -1 \Rightarrow (-\infty, 0)$ ...(*i*)

$$B = \left\{ x \in R : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r} \right)^{x-3} < 3^{-3x} \right\}$$

$$\Rightarrow 3^x \left(\frac{3 \cdot \frac{1}{10}}{1 - \frac{1}{10}} \right)^{x-3} < 3^{-3x}$$

$$\Rightarrow \left(\frac{1}{9}\right)^{x-3} < 3^{-5x+3}$$

$\Rightarrow 3^{6-2x} < 3^{3-5x}$

$\Rightarrow 6 - 2x < 3 - 5x$

$\Rightarrow -3x > 3$

$\Rightarrow x < -1$ ...(*ii*)

From equ (*i*) & eqn (*ii*), we get

Hence, $B \subset A$

[6 April, 2023 (Shift-I)]

53. (3)

Given 6 boys and n girls

Total ways of farming team of 3

Members under given condition

$= {}^6C_1 . {}^nC_2 + {}^6C_2 . {}^nC_1$

$\Rightarrow {}^6C_1 . {}^nC_2 + {}^6C_2 . {}^nC_1 = 1440$

$n^2 + 4n = 20 \times 24$

$\Rightarrow \quad n(n+4) = 20 \times 24$

$\Rightarrow \quad n = 20$

54. (3)

Let the other root is α

Clearly $2 < \alpha < 3$

$\therefore \quad 1 < \alpha + (-1) < 2$

[28 June, 2022 (Shift-II)]

55. (2)

$$\frac{\sum_{i=1}^{n} 3^{y_i}}{n} \geq \left(\prod_{i=1}^{n} 3^{y_i} \right)^{1/n}$$

$$\therefore \quad \log_3 \sum_{i=1}^{n} 3^{y_i} = \log_3 n + \frac{9}{n}$$

Similarly

$$\frac{\sum x_i}{n} \geq \left(\Pi x_i \right)^{1/n}$$

$$\log_3 \frac{9}{n} \geq \frac{1}{n} \log_3 \left(\Pi x_i \right)$$

$$\therefore \quad M = n \log_3 \left(\frac{9}{n} \right)$$

[JEE Adv, 2020]

56. (3)

$$\frac{\left(\alpha^{23} + \beta^{23} \right)(\alpha + \beta)}{\left(\alpha^{23} + \beta^{23} \right)} = \alpha + \beta = -\sqrt{6}$$

[12 April, 2023 (Shift-I)]

57. (3)

Mean = 200 and let common difference = d

First term given = 2

$$\text{Mean} = \frac{\frac{100}{2}\left(2 \times 2 + 99d\right)}{100} = 200$$

$\Rightarrow 4 + 99d = 400$

$\Rightarrow d = 4$

And $y_i = i(x_i - i)$

$= i(2 + (i-1)4 - i) = 3i^2 - 2i$

$$\text{Mean} = \frac{\sum y_i}{100}$$

$$= \frac{1}{100} \sum_{i=1}^{100} 3i^2 - 2i$$

$$= \frac{1}{100} \left\{ \frac{3 \times 100 \times 101 \times 201}{6} - \frac{2 \times 100 \times 101}{2} \right\}$$

$$= 101 \left\{ \frac{201}{2} - 1 \right\} = 101 \times 99.5$$

$= 10049 \cdot 50$

[11 April, 2023 (Shift-I)]

58. (2)

4×3^{19}

$a_1 a_2 a_3 a_4 = a_1^4 r^6 = 2^4 \times 3^6 \quad \ldots..(i)$

$a_n \cdot a_{n-1} \cdot a_{n-2} \cdot a_{n-3}$

$= a_1^4 \cdot r^{4n-10} = 2^4 \times 3^{70} \quad \ldots.(ii)$

$\Rightarrow \quad r^{4n-16} = 3^{64}$

$\Rightarrow \quad r = 3, \; n = 20$

[1 Feb, 2023 (Shift-I)]

59. (1)

Consider, and $z_1 = x_1 + iy_1 \;\; z_2 = x_2 + iy_2$

$\text{Im}(z_1 z_2)$

$= x_1 y_2 + x_2 y_1 = 0$

$$\Rightarrow \quad \frac{x_1}{x_2} = -\frac{y_1}{y_2} > 0 \qquad \ldots.(i)$$

$\text{Im}(z_1 + z_2) = 0$

$\Rightarrow \quad y_1 + y_2 = 0 \qquad \ldots.(ii)$

$\Rightarrow \quad y_1 y_2 < 0 \qquad \ldots..(iii)$

(i) & (iii) gives result $x_1 = x_2$

Either $x_1, x_2 > 0$

Or $x_1, x_2 < 0$

[29 Jan, 2023 (Shift-I)]

60. (4)

$$\text{Let } z = \frac{3 + 2i \sin\theta}{4 - i \sin\theta} \times \frac{4 + i \sin\theta}{4 + i \sin\theta}$$

$$\text{Im}(z) = \frac{(11 \sin\theta)}{16 + \sin^2\theta}$$

$\Rightarrow \quad 11 \sin\theta = 0$ (for purely real)

$\therefore \quad \theta = 0, \pi, 2\pi$

[8 Jan, 2023 (Shift-II)]

61. (3)

In $\left(\frac{x^{\frac{5}{2}}}{2} - \frac{4}{x^l} \right)^9$ general term T_{r+1} is

$$T_{r+1} = {}^9C_r \frac{\left(x^{5/2} \right)^{9-r}}{2^{9-r}} \left(\frac{-4}{x^l} \right)^r$$

$$= 45 - 5r - 2lr = 0$$

$$r = \frac{45}{5 + 2l} \qquad ..(i)$$

Now, according to the question,

$$(-1)^r \frac{{}^9C_r}{2^{9-r}} 4^r = -84$$

$$= (-1)^r \, {}^9C_r \, 2^{3r-9} = -21 \times 4$$

Only natural value of r possible if $3r - 9 = 0$

$\Rightarrow r = 3 \Rightarrow$ and ${}^9C_3 = 84$

$\therefore l = 5$ from equation (i)

Now, coefficient of $x^{-3l} = x^{\frac{45}{2} - \frac{5r}{2} - lr}$ at $l = 5$,

gives $r = 5$

$\therefore {}^9C_5(-1)\dfrac{4^5}{2^4} = 2^\alpha \times \beta$

$= -63 \times 2^7$

compare with $\alpha \times \beta$

$\Rightarrow \alpha = 7, \beta = -63$

[31 Jan, 2023 (Shift-II)]

62. (4)

$\left(3+2\sqrt{2}\right)^{17} = {}^{17}C_0 3^{17} + {}^{17}C_1 3^{16}\left(2\sqrt{2}\right) + ... = I + f$

$\left(3-2\sqrt{2}\right)^{17} = {}^{17}C_0 3^{17} - {}^{17}C_1 3^{16}\left(2\sqrt{2}\right) + ... = f'$

Adding both we get

$\left(3-2\sqrt{2}\right)^{17} + \left(3-2\sqrt{2}\right)^{17}$

$= 2\left({}^{17}C_0 + {}^{17}C_2 3^{15}\left(2\sqrt{2}\right)^2 + ...\right)$

$= 2\left({}^{17}C_0 + {}^{17}C_2 3^{15}\left(2\sqrt{2}\right)^2 + ...\right) = I + f + f'$

as $f + f' = 1\left(\therefore 0 < \left(3-2\sqrt{17}\right)^{17} < 1\right)$

$I + 1$ = Even integer

$\Rightarrow$ I = odd integer $= \left[\left(3+2\sqrt{2}\right)^{17}\right]$

[30 Jan, 2023 (Shift-II)]

63. (3)

$M = 25$

$x(70-x) \geq 24 \times 25$

$10 \leq x \leq 60$

{12,15,18, ... 54, 57, 60}

$P(A) = \dfrac{34}{51} = \dfrac{2}{3}$

[25 Jan, 2023 (Shift-I)]

64. (1)

$\lim\limits_{x\to 0}\left(\dfrac{(\sin x + \sin 3x)}{x}\dfrac{x^3}{4\sin^3 x}\right) = \dfrac{4}{4} = 1$

[24 Jan, 2023 (Shift-I)]

65. (2)

$64\cos\dfrac{\pi}{33}\cos\dfrac{2\pi}{33}\cos\dfrac{2^2\pi}{33}\cos\dfrac{2^3\pi}{33}\cos\dfrac{2^4\pi}{33}$

$\because \cos A\cos 2A\cos 2^2 A\cos 2^3 A \ldots \cos 2^{n-1}$

$= \dfrac{\sin\left(2^n A\right)}{2^n \sin A}$

Here,

$A = \dfrac{\pi}{33}, n = 5$

$\dfrac{64\sin\left(2^5\dfrac{\pi}{33}\right)}{2^5\sin\left(\dfrac{\pi}{33}\right)} = \dfrac{64\sin\left(\dfrac{32\pi}{33}\right)}{32\sin\left(\dfrac{\pi}{33}\right)}$

$= \dfrac{2\sin\left(\pi - \dfrac{\pi}{33}\right)}{\sin\left(\dfrac{\pi}{33}\right)} = 2$

[10 April, 2023 (Shift-I)]

66. (4)

We have

$2\cos\dfrac{x+y}{2}\cos\dfrac{x-y}{2} - \left(2\cos^2\left(\dfrac{x+y}{2}\right) - 1\right) = \dfrac{3}{2}$

$\Rightarrow \quad 2\cos\left(\dfrac{x+y}{2}\right)\left(\cos\dfrac{x-y}{2} - \cos\dfrac{x+y}{2}\right) = \dfrac{1}{2}$

$\Rightarrow \quad 2\cos\dfrac{x+y}{2}\cdot 2\sin\dfrac{x}{2}.\sin\dfrac{y}{2} = \dfrac{1}{2}$

$\Rightarrow \quad \left(\cos\dfrac{x}{2}\cos\dfrac{y}{2} - \sin\dfrac{x}{2}\sin\dfrac{y}{2}\right)\times\sin\dfrac{x}{2}\sin\dfrac{y}{2} = \dfrac{1}{8}$

$\Rightarrow \left(\sin\dfrac{x}{2}\sin\dfrac{y}{2}\right)^2 - \left(\cos\dfrac{x}{2}\cos\dfrac{y}{2}\right)\left(\sin\dfrac{x}{2}\sin\dfrac{y}{2}\right) + \dfrac{1}{8} = 0$

$D \geq 0$

$\Rightarrow \quad \cos\dfrac{x}{2}\cos\dfrac{y}{2} \geq \dfrac{1}{\sqrt{2}}$

[25 Feb, 2021 (Shift-II)]

67. (3)

Case I: For log to be defined

$x + 2 > 0 \Rightarrow x > -2 \Rightarrow x \in (-2, \infty)$

Case II: $x + 1 > 0 \Rightarrow x > -1$

And $x + 1 \neq 1 \Rightarrow x \neq 0 \Rightarrow x \in (-1, 0) \cup (0, \infty)$

Case III: $\Rightarrow x \in (0, \infty)$

Case IV: Denominator:

$\Rightarrow (x-3)(x+1) \neq 0$

$\Rightarrow x \neq -1, 3$

Taking intersection of these cases, we get $(0, \infty) - \{3\}$

[29 Jan, 2023 (Shift-I)]

68. (1)

Only possibility $\alpha = 0$, $\beta = 1$

Equation of circle by figure with AB as diameter

$(x - 0)(x - 1) + (y - 0)(y - 1) = 0$

Image of center of circle $\left(\dfrac{1}{2}, \dfrac{1}{2}\right)$ in $y = -x$ is

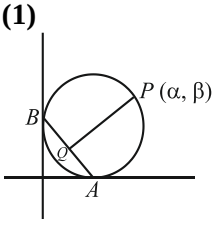
$\left(-\dfrac{1}{2}, -\dfrac{1}{2}\right)$

and radius will be same.

So, $x^2 + y^2 + x + y = 0$

[25 Jan, 2023 (Shift-I)]

69. (1)



Let equation of circle is $(x-a)^2+(y-a)^2=a^2$

Which is passing through $P(\alpha,\beta)$

Then $(\alpha-a)^2+(\beta-a)^2=a^2$

$\alpha^2+\beta^2-2\alpha a-2\beta a+a^2=0$

Here equation of AB is $x+y=a$

Let $Q(\alpha',\beta')$ be foot of perpendicular of P on AB

$$\frac{\alpha'-\alpha}{1}=\frac{\beta'-\beta}{1}=\frac{-(\alpha+\beta-a)}{2}$$

$$PQ^2=(\alpha'-\alpha)^2+(\beta'-\beta)^2$$

$$=\frac{1}{4}(\alpha+\beta-a)^2+\frac{1}{4}(\alpha+\beta-a)^2$$

$$\Rightarrow 144=\frac{1}{2}(\alpha+\beta-a)^2$$

$$\Rightarrow 288=\alpha^2+\beta^2-2\alpha a-2\beta a+a^2+2\alpha\beta$$

$$\Rightarrow 288=2\alpha\beta$$

$$\Rightarrow \alpha\beta=144$$

[6 April, 2023 (Shift-I)]

70. (3)

Circle $(x-2)^2+y^2=r^2$

Solve with ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$

$$\frac{x^2}{36}+\frac{r^2-(x-2)^2}{16}=1$$

$5x^2-36x+180-9r^2=0$

$D=0$

$$r=\frac{8}{\sqrt{5}}$$

[24 Jan, 2023 (Shift-I)]

71. (14)

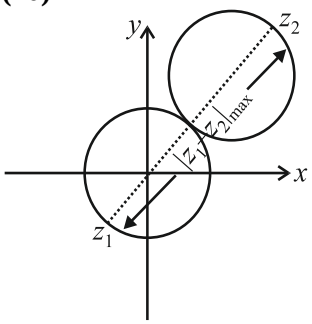
Given $2a=8\sqrt{2}$

$$b^2=a^2\left(\frac{11}{4}-1\right), 4b^2=7.64.2$$

$2b=4\sqrt{14}$

[29 June, 2022 (Shift-I)]

72. (10)



[12 Jan, 2019 (Shift-II)]

73. (3)

$2^m+2^n=288$

$\therefore (m,n)=(8,5)$ *i.e.* $|m-n|=3$

[6 Sep. 2020 (Shift-I)]

74. (25)

Given, $\bar{X}=\frac{\sum_{i=1}^{25}x_i}{N}=42$

$$\Rightarrow \sum_{i=1}^{25}x_i=42N$$

$$\sum_{i=1}^{24}x_i=42N-60+35=41N$$

$N=25$

[18 March, 2021(Shift-I)]

75. (135)

Required numbers

=(when zero is present) + (when zero is not present) – (numbers formed by one digit only)

$=9\times 2^4+{}^9C_2\cdot 2^5-(9+2\,{}^9C_2)=9\times 135$

[24 Jan, 2023 (Shift-II)]

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